

The Complete Treatise on the Four-Sector Radical-Balance Interest Rate

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Abstract

This paper develops a mathematical and interdisciplinary framework for a four-sector radical-balance interest-rate system. The primitive relation is

$$\sqrt{a(t) - r(t)} + \sqrt{b(t) - r(t)} = c(t), \quad (1)$$

together with the three sign variants generated by replacing the coefficients of the radicals with the configurations

$$(+1, -1), \quad (-1, +1), \quad (-1, -1). \quad (2)$$

The four-sector system is represented as an action of the sign group $\{\pm 1\}^2$ on the radical coordinates. A central result is that all four algebraic sectors generate the same branch-invariant interest-rate expression,

$$r(t) = \frac{a(t) + b(t)}{2} - \frac{c(t)^2}{4} - \frac{[a(t) - b(t)]^2}{4c(t)^2}, \quad (3)$$

while the principal-square-root restrictions determine which sector is actually feasible. The paper establishes feasibility, symmetry, extremal, identification, perturbation, spectral, and arbitrage results. It then constructs an econometric framework for estimating the rate, discusses nonlinear frequency-domain effects, and develops interdisciplinary interpretations in economics, finance, statistics, physics, chemistry, biology, psychology, philosophy, religion, political science, geopolitics, diplomacy, international relations, welfare economics, warfare economics, computer science, algorithms, data science, and artificial intelligence. The interdisciplinary sections distinguish mathematical consequences from analogical interpretations and empirical hypotheses.

The paper ends with “The End”

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1 Introduction

Consider an interest-rate process $r(t)$ constrained by two radical distances from time-dependent anchors $a(t)$ and $b(t)$:

$$\sqrt{a(t) - r(t)} + \sqrt{b(t) - r(t)} = c(t). \quad (4)$$

The equation is nonlinear because the interest rate appears inside two square-root operators. It is also structurally symmetric in $a(t)$ and $b(t)$. The purpose of this paper is to investigate the complete family obtained by allowing independent signs before the two radicals.

The four equations are

$$\sqrt{a - r} + \sqrt{b - r} = c, \quad (5)$$

$$\sqrt{a - r} - \sqrt{b - r} = c, \quad (6)$$

$$-\sqrt{a - r} + \sqrt{b - r} = c, \quad (7)$$

$$-\sqrt{a - r} - \sqrt{b - r} = c. \quad (8)$$

The principal mathematical question is whether these equations define four different interest rates. The answer is subtle. Algebraically, they produce the same expression for r , but they correspond to different sign sectors of the radical coordinates. Hence the interest rate is branch invariant while the radical representation is branch dependent.

This distinction motivates the central concept of the paper.

Definition 1.1 (Four-Sector Radical-Balance System). *The four-sector radical-balance system is the family*

$$s_1 \sqrt{a(t) - r(t)} + s_2 \sqrt{b(t) - r(t)} = c(t), \quad (9)$$

where

$$(s_1, s_2) \in \{(+1, +1), (+1, -1), (-1, +1), (-1, -1)\}. \quad (10)$$

The paper subsequently distinguishes the purely algebraic system from an economic model in which the four sectors possess different economic states, prices, funding costs, or institutional constraints. This distinction is essential. The exact algebraic system is arbitrage neutral because all four sectors share the same $r(t)$. Arbitrage arises only after economically meaningful sectoral deviations are introduced.

2 Primitive Mathematical Structure

2.1 Radical coordinates

Define

$$x(t) = \sqrt{a(t) - r(t)}, \quad y(t) = \sqrt{b(t) - r(t)}. \quad (11)$$

For principal square roots,

$$x(t) \geq 0, \quad y(t) \geq 0. \quad (12)$$

The system becomes

$$s_1 x(t) + s_2 y(t) = c(t). \quad (13)$$

Suppressing the time argument temporarily,

$$x^2 = a - r, \quad y^2 = b - r. \quad (14)$$

Consequently,

$$x^2 - y^2 = a - b. \quad (15)$$

Factorization yields

$$(x - y)(x + y) = a - b. \quad (16)$$

The radical system is therefore simultaneously a nonlinear equation in r and a two-coordinate geometric system in x and y .

2.2 The sign-sector representation

Define the sector vector

$$\mathbf{s} = \begin{pmatrix} s_1 \\ s_2 \end{pmatrix}, \quad s_1, s_2 \in \{-1, +1\}, \quad (17)$$

and the radical vector

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}. \quad (18)$$

Then

$$\mathbf{s}^\top \mathbf{x} = c. \quad (19)$$

The four sectors are therefore

$$\mathcal{S} = \{(+, +), (+, -), (-, +), (-, -)\}. \quad (20)$$

This is naturally associated with the group

$$\mathcal{G} = \mathbb{Z}_2 \times \mathbb{Z}_2, \quad (21)$$

acting on the signs of the radical coordinates.

3 The Branch-Invariant Interest Rate

Theorem 3.1 (Branch Invariance). *Suppose $c \neq 0$. Every sector of the four-sector radical-balance system generates the same algebraic value of r :*

$$r = \frac{a+b}{2} - \frac{c^2}{4} - \frac{(a-b)^2}{4c^2}. \quad (22)$$

Proof. Let

$$x = \sqrt{a-r}, \quad y = \sqrt{b-r}. \quad (23)$$

The sector equation is

$$s_1x + s_2y = c. \quad (24)$$

Squaring gives

$$x^2 + y^2 + 2s_1s_2xy = c^2. \quad (25)$$

Because

$$x^2 = a-r, \quad y^2 = b-r, \quad (26)$$

we have

$$a+b-2r+2s_1s_2xy = c^2. \quad (27)$$

Also,

$$(x^2 - y^2)^2 = (a-b)^2. \quad (28)$$

The identity

$$(x^2 - y^2)^2 = (x^2 + y^2)^2 - 4x^2y^2 \quad (29)$$

combined with the sector equation yields

$$r = \frac{a+b}{2} - \frac{1}{4} \left(c^2 + \frac{(a-b)^2}{c^2} \right). \quad (30)$$

The sector signs have disappeared. Hence the value of r is algebraically invariant across the four sectors. $\square$

The time-dependent result is therefore

$$r(t) = \frac{a(t)+b(t)}{2} - \frac{c(t)^2}{4} - \frac{[a(t)-b(t)]^2}{4c(t)^2}. \quad (31)$$

4 Reconstruction of the Radical Coordinates

Theorem 4.1 (Sector-Dependent Radical Coordinates). *For a sector (s_1, s_2) , the radical coordinates are*

$$x = \frac{1}{2} \left(s_1c + s_1s_2 \frac{a-b}{c} \right), \quad (32)$$

$$y = \frac{1}{2} \left(s_2c - s_1s_2 \frac{a-b}{c} \right). \quad (33)$$

Proof. From

$$s_1x + s_2y = c \quad (34)$$

and

$$x^2 - y^2 = a - b, \quad (35)$$

we obtain

$$(s_1x - s_2y)(s_1x + s_2y) = a - b. \quad (36)$$

Therefore

$$s_1x - s_2y = \frac{a-b}{c}. \quad (37)$$

Adding the two equations gives

$$2s_1x = c + \frac{a-b}{c}. \quad (38)$$

Subtracting gives

$$2s_2y = c - \frac{a-b}{c}. \quad (39)$$

The stated expressions follow. $\square$

Thus the four sectors are different coordinate representations of the same branch-invariant rate.

5 Feasibility and Sector Geometry

Because x and y are principal square roots, feasibility requires

$$x \geq 0, \quad y \geq 0. \quad (40)$$

Theorem 5.1 (Feasibility Conditions). *For a sector (s_1, s_2) , feasibility requires*

$$s_1c + s_1s_2\frac{a-b}{c} \geq 0, \quad (41)$$

$$s_2c - s_1s_2\frac{a-b}{c} \geq 0. \quad (42)$$

For $c > 0$, the sector conditions simplify to the following table.

Table 1: Principal-root feasibility by sector for $c > 0$

Sector	Equation	Feasibility condition
++	$x + y = c$	$c^2 \geq a - b $
+-	$x - y = c$	$a - b \geq c^2$
-+	$-x + y = c$	$b - a \geq c^2$
--	$-x - y = c$	impossible

For $c < 0$, the sign orientation is reversed. Hence the algebraic formula for r should always be accompanied by its principal-root feasibility conditions.

Theorem 5.2 (Sector Exclusivity). *For $c > 0$, at most one of the ++, +-, and -+ sectors is strictly feasible.*

Proof. The +- condition requires

$$a - b > 0 \quad (43)$$

and

$$a - b \geq c^2. \quad (44)$$

The -+ condition requires

$$b - a > 0 \quad (45)$$

and

$$b - a \geq c^2. \quad (46)$$

These cannot simultaneously hold. The ++ condition requires

$$|a - b| \leq c^2, \quad (47)$$

which is incompatible with the strict interior of either difference sector. Only boundary cases can produce simultaneous representations involving a zero radical. $\square$

6 Bounds and Extremal Properties

Proposition 6.1 (Interest-Rate Upper Bound). *Every feasible solution satisfies*

$$\boxed{r \leq \min(a, b).} \quad (48)$$

Proof. Since

$$a - r = x^2 \geq 0 \quad (49)$$

and

$$b - r = y^2 \geq 0, \quad (50)$$

we have

$$r \leq a, \quad r \leq b. \quad (51)$$

Hence

$$r \leq \min(a, b). \quad (52)$$

□

The two radical gaps are explicitly

$$a - r = \frac{1}{4} \left(c + \frac{a - b}{c} \right)^2, \quad (53)$$

$$b - r = \frac{1}{4} \left(c - \frac{a - b}{c} \right)^2. \quad (54)$$

Theorem 6.2 (Extremal Rate). *For fixed a and b , the rate*

$$r(c) = \frac{a + b}{2} - \frac{1}{4} \left(c^2 + \frac{(a - b)^2}{c^2} \right) \quad (55)$$

is maximized at

$$\boxed{c^2 = |a - b|.} \quad (56)$$

The maximum is

$$\boxed{r_{\max} = \min(a, b).} \quad (57)$$

Proof. Let

$$z = c^2 > 0. \quad (58)$$

Then

$$r(z) = \frac{a + b}{2} - \frac{1}{4} \left(z + \frac{(a - b)^2}{z} \right). \quad (59)$$

Differentiation gives

$$\frac{dr}{dz} = -\frac{1}{4} \left(1 - \frac{(a - b)^2}{z^2} \right). \quad (60)$$

The stationary point satisfies

$$z^2 = (a - b)^2. \quad (61)$$

Since $z > 0$,

$$z = |a - b|. \quad (62)$$

Substitution yields

$$r_{\max} = \frac{a + b - |a - b|}{2} = \min(a, b). \quad (63)$$

□

7 Symmetry

Theorem 7.1 (Anchor Symmetry). *The branch-invariant rate satisfies*

$$r(a, b, c) = r(b, a, c). \quad (64)$$

This follows because the rate depends on $a - b$ only through

$$(a - b)^2. \quad (65)$$

However, the sector labels transform according to

$$(a, b, +, -) \longleftrightarrow (b, a, -, +). \quad (66)$$

Hence the $+-$ and $-+$ sectors form a symmetry pair.

8 The Equal-Anchor Limit

Suppose

$$a(t) = b(t) = q(t). \quad (67)$$

Then

$$r(t) = q(t) - \frac{c(t)^2}{4}. \quad (68)$$

The original sector equation becomes

$$2\sqrt{q(t) - r(t)} = c(t), \quad (69)$$

and therefore

$$\sqrt{q(t) - r(t)} = \frac{|c(t)|}{2}. \quad (70)$$

This is the maximally symmetric configuration.

9 Frequency-Domain Representation

The nonlinear structure has important spectral implications.

Let the Fourier transform be defined by

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt. \quad (71)$$

The exact rate equation is

$$r(t) = \frac{a(t) + b(t)}{2} - \frac{c(t)^2}{4} - \frac{[a(t) - b(t)]^2}{4c(t)^2}. \quad (72)$$

Therefore

$$R(\omega) = \frac{A(\omega) + B(\omega)}{2} - \frac{1}{4} \mathcal{F}\{c^2\}(\omega) - \frac{1}{4} \mathcal{F}\left\{\frac{(a-b)^2}{c^2}\right\}(\omega). \quad (73)$$

Using convolution,

$$\mathcal{F}\{c^2\}(\omega) = \frac{1}{2\pi} (C * C)(\omega). \quad (74)$$

Consequently,

$$\boxed{R(\omega) = \frac{A(\omega) + B(\omega)}{2} - \frac{1}{8\pi} (C * C)(\omega) - \frac{1}{4} \mathcal{F}\left\{\frac{(a-b)^2}{c^2}\right\}(\omega).} \quad (75)$$

Importantly,

$$\mathcal{F}\{\sqrt{a-r}\} \neq \sqrt{A-R} \quad (76)$$

in general. Fourier transformation does not commute with nonlinear square-root operations.

9.1 Symmetric spectral case

If

$$a(t) = b(t) = q(t), \quad (77)$$

then

$$r(t) = q(t) - \frac{c(t)^2}{4}, \quad (78)$$

and therefore

$$\boxed{R(\omega) = Q(\omega) - \frac{1}{8\pi}(C * C)(\omega).} \quad (79)$$

Thus the model is intrinsically a nonlinear frequency-mixing system.

10 Harmonic Generation

Consider

$$c(t) = c_0 + A \cos(\omega_0 t). \quad (80)$$

Then

$$c(t)^2 = c_0^2 + 2c_0 A \cos(\omega_0 t) + A^2 \cos^2(\omega_0 t) \quad (81)$$

$$= c_0^2 + \frac{A^2}{2} + 2c_0 A \cos(\omega_0 t) + \frac{A^2}{2} \cos(2\omega_0 t). \quad (82)$$

Consequently, the interest rate contains the frequencies

$$0, \quad \omega_0, \quad 2\omega_0. \quad (83)$$

This is the simplest demonstration that nonlinear radical-balance dynamics can generate harmonics not present in the original input.

11 Local Linearization

Let

$$a(t) = a_0 + \delta a(t), \quad (84)$$

$$b(t) = b_0 + \delta b(t), \quad (85)$$

$$c(t) = c_0 + \delta c(t), \quad (86)$$

$$r(t) = r_0 + \delta r(t), \quad (87)$$

where

$$\sqrt{a_0 - r_0} + \sqrt{b_0 - r_0} = c_0. \quad (88)$$

A first-order expansion gives

$$\frac{\delta a - \delta r}{2\sqrt{a_0 - r_0}} + \frac{\delta b - \delta r}{2\sqrt{b_0 - r_0}} = \delta c. \quad (89)$$

Define

$$D = \frac{1}{\sqrt{a_0 - r_0}} + \frac{1}{\sqrt{b_0 - r_0}}. \quad (90)$$

Then

$$\delta r = \frac{1}{D} \left[\frac{\delta a}{\sqrt{a_0 - r_0}} + \frac{\delta b}{\sqrt{b_0 - r_0}} - 2\delta c \right]. \quad (91)$$

Taking Fourier transforms gives

$$R_1(\omega) = \alpha A(\omega) + \beta B(\omega) - \gamma C(\omega), \quad (92)$$

where

$$\alpha = \frac{1/\sqrt{a_0 - r_0}}{D}, \quad (93)$$

$$\beta = \frac{1/\sqrt{b_0 - r_0}}{D}, \quad (94)$$

$$\gamma = \frac{2}{D}. \quad (95)$$

Furthermore,

$$\alpha + \beta = 1. \quad (96)$$

Thus the local system behaves like a linear spectral transfer function, even though the global system is nonlinear.

12 Econometric Identification and Estimation

12.1 Structural identification

If $a(t)$, $b(t)$, and $c(t)$ are observed without error, then $r(t)$ is structurally identified:

$$\hat{r}_t^{\text{struct}} = \frac{\hat{a}_t + \hat{b}_t}{2} - \frac{\hat{c}_t^2}{4} - \frac{(\hat{a}_t - \hat{b}_t)^2}{4\hat{c}_t^2}. \quad (97)$$

This is a plug-in estimator rather than an econometric estimator of an unknown structural parameter.

12.2 Measurement-error specification

A stochastic observation equation can be written

$$c_t = \sqrt{a_t - r_t} + \sqrt{b_t - r_t} + \varepsilon_t. \quad (98)$$

A nonlinear least-squares estimator is

$$\hat{\mathbf{r}} = \arg \min_{\mathbf{r}} \sum_{t=1}^T \left[\sqrt{a_t - r_t} + \sqrt{b_t - r_t} - c_t \right]^2, \quad (99)$$

subject to

$$r_t < \min(a_t, b_t). \quad (100)$$

A dynamic specification may instead be

$$r_t = \rho r_{t-1} + (1 - \rho)\mu + \eta_t. \quad (101)$$

Together with the nonlinear observation equation, this creates a nonlinear state-space model. Potential estimators include

$$\text{maximum likelihood,} \quad \text{particle filtering,} \quad \text{Bayesian state-space estimation.} \quad (102)$$

12.3 Numerical instability

The structural estimator contains

$$\frac{(a_t - b_t)^2}{4c_t^2}. \quad (103)$$

Consequently, small values of c_t can produce substantial numerical sensitivity.

The derivative with respect to c is

$$\frac{\partial r}{\partial c} = -\frac{c}{2} + \frac{(a - b)^2}{2c^3}. \quad (104)$$

Hence

$$\left| \frac{\partial r}{\partial c} \right| \rightarrow \infty \quad \text{as} \quad c \rightarrow 0, \quad (105)$$

unless $a = b$.

This is an important empirical identification issue.

13 The Four-Sector Arbitrage Extension

The exact four-sector radical system satisfies

$$r_{++} = r_{+-} = r_{-+} = r_{--} = r. \quad (106)$$

Therefore,

$$r_i - r_j = 0 \quad (107)$$

for all sectors i, j .

The exact mathematical system is consequently arbitrage neutral.

To model actual financial sectors, introduce sectoral deviations:

$$r_i(t) = r(t) + \varepsilon_i(t), \quad i \in \{1, 2, 3, 4\}. \quad (108)$$

Then

$$r_i(t) - r_j(t) = \varepsilon_i(t) - \varepsilon_j(t). \quad (109)$$

Definition 13.1 (Sectoral Arbitrage Pressure). *The friction-adjusted arbitrage pressure from sector j toward sector i is*

$$A_{ij}(t) = \lambda_{ij} [r_i(t) - r_j(t) - \kappa_{ij}(t)]_+, \quad (110)$$

where $\lambda_{ij} > 0$ is a flow coefficient and $\kappa_{ij}(t)$ represents transaction costs and other relevant wedges.

The positive-part operator is

$$[x]_+ = \max(x, 0). \quad (111)$$

14 The Three-Against-One Configuration

Suppose one sector has the lowest rate:

$$r_L(t) = \min_i r_i(t). \quad (112)$$

The aggregate arbitrage pressure directed against the low-rate sector is

$$\boxed{\mathcal{A}_L(t) = \sum_{i \neq L} \lambda_{iL} [r_i(t) - r_L(t) - \kappa_{iL}(t)]_+}. \quad (113)$$

If three other sectors have rates exceeding the low-rate sector by more than their respective frictional wedges, then all three corresponding terms are positive.

This produces the mathematical structure

$$\boxed{\text{three high-rate sectors} \longrightarrow \text{one low-rate sector.}} \quad (114)$$

The terminology “attack” is interpreted here as a metaphor for arbitrage-induced capital-flow pressure rather than as a literal military action.

15 No-Arbitrage Equilibrium

Theorem 15.1 (Four-Sector Arbitrage Neutrality). *If*

$$r_i(t) = r(t) \quad (115)$$

for all four sectors and all relevant arbitrage frictions satisfy

$$\kappa_{ij}(t) \geq 0, \quad (116)$$

then

$$A_{ij}(t) = 0 \quad (117)$$

for every pair i, j .

Proof. Since

$$r_i - r_j = 0, \quad (118)$$

we have

$$r_i - r_j - \kappa_{ij} = -\kappa_{ij} \leq 0. \quad (119)$$

Therefore

$$[r_i - r_j - \kappa_{ij}]_+ = 0, \quad (120)$$

which implies

$$A_{ij} = 0. \quad (121)$$

□

Thus the radical-balance solution can be interpreted as a benchmark no-arbitrage manifold.

16 Financial Interpretation

The four-sector model can be interpreted as a stylized financial system in which the same structural rate is represented through four sign sectors.

Possible interpretations of $a(t)$ and $b(t)$ include

$$\begin{array}{cccc} \textit{benchmark} & \textit{discount - rate} & \textit{funding} & \textit{risk - adjusted} \\ \textit{yields} & \textit{anchors} & \textit{rates} & \textit{hurdle rates} \end{array}, \quad (122)$$

while $c(t)$ may represent an aggregate radical spread or nonlinear constraint.

These interpretations are not imposed by the mathematics. They constitute candidate economic mappings.

A sectoral extension can include

$$r_{i,t} = r_t + \varepsilon_{i,t}, \quad (123)$$

where $\varepsilon_{i,t}$ captures liquidity, credit, regulatory, institutional, or market-segmentation effects.

17 Welfare Economics

A stylized welfare function may be written

$$W(t) = \sum_{i=1}^4 \omega_i U_i(C_i(t)), \quad (124)$$

where

$$\omega_i \geq 0, \quad \sum_{i=1}^4 \omega_i = 1. \quad (125)$$

Interest-rate differentials influence intertemporal allocation through the Euler condition. For a representative agent,

$$U'(C_t) = \beta(1 + r_t) \mathbb{E}_t[U'(C_{t+1})]. \quad (126)$$

In a multi-sector economy, different r_i 's can therefore create different intertemporal incentives.

However, welfare maximization cannot be inferred from arbitrage profitability alone. Redistribution, externalities, incomplete markets, market power, and risk can make the welfare implications of a rate differential nontrivial.

18 Macroeconomic Interpretation

The radical-balance system can be embedded in a macroeconomic state vector

$$\mathbf{z}_t = \begin{pmatrix} r_t \\ \pi_t \\ y_t \\ m_t \\ q_t \end{pmatrix}, \quad (127)$$

where the components may represent the interest rate, inflation, output, money, and an asset-price state.

A reduced-form transition equation can be written

$$\mathbf{z}_{t+1} = F(\mathbf{z}_t, \mathbf{u}_t) + \boldsymbol{\eta}_t. \quad (128)$$

The radical-balance relation then functions as a nonlinear structural constraint

$$G(\mathbf{z}_t) = 0. \quad (129)$$

This creates a constrained nonlinear dynamical system.

19 Statistics and Uncertainty

Suppose

$$\hat{\boldsymbol{\theta}} = \begin{pmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{pmatrix} \quad (130)$$

has covariance matrix

$$\hat{\Sigma}_{\boldsymbol{\theta}}. \quad (131)$$

Let

$$r = g(a, b, c). \quad (132)$$

The delta method gives

$$\text{Var}(\hat{r}) \approx \nabla g^\top \hat{\Sigma}_\theta \nabla g. \quad (133)$$

The gradient is

$$\frac{\partial r}{\partial a} = \frac{1}{2} - \frac{a-b}{2c^2}, \quad (134)$$

$$\frac{\partial r}{\partial b} = \frac{1}{2} + \frac{a-b}{2c^2}, \quad (135)$$

$$\frac{\partial r}{\partial c} = -\frac{c}{2} + \frac{(a-b)^2}{2c^3}. \quad (136)$$

This allows standard errors for the structural rate estimate to be constructed.

20 Physics Analogy

The mathematical structure has an analogy with a system possessing multiple coordinate sectors but a common invariant.

The sector transformation

$$(x, y) \mapsto (s_1 x, s_2 y) \quad (137)$$

resembles a discrete symmetry operation. The scalar r is invariant under this operation because it depends on quantities such as

$$x^2, \quad y^2, \quad (x^2 - y^2)^2. \quad (138)$$

This is a mathematical symmetry analogy, not a claim that the interest-rate system is a physical gauge theory.

21 Chemistry Analogy

The four sectors can be viewed heuristically as four sign configurations of two interacting radical coordinates.

The analogy to chemical reaction networks is limited but suggestive: different microscopic configurations can sometimes produce the same macroscopic observable.

A generic reaction-network representation is



with an observable equilibrium state that need not uniquely identify the microscopic pathway.

In the present model, the analogous distinction is between

$$\text{sector representation} \quad (140)$$

and

$$\text{invariant rate}. \quad (141)$$

22 Biological Systems Analogy

A biological network may contain multiple microscopic pathways leading to a common macroscopic phenotype. A stylized dynamical representation is

$$\dot{\mathbf{x}} = F(\mathbf{x}, \mathbf{u}). \quad (142)$$

The radical-balance model similarly separates a representation layer from an invariant state variable. This analogy suggests a possible empirical research hypothesis:

$$\text{different sectoral mechanisms} \longrightarrow \text{common equilibrium observable}. \quad (143)$$

The hypothesis requires empirical validation and is not a theorem of biology.

23 Psychology and Decision Theory

In behavioral finance, agents may respond differently to the same objective rate because of heterogeneous beliefs, loss aversion, ambiguity aversion, or bounded rationality.

A generic utility function can be represented as

$$U(c) = \mathbb{E}[V(c)] - \lambda \mathbb{E}[V(c) - V(c_{\text{reference}})]_+, \quad (144)$$

where the second term represents a stylized downside-sensitive component.

Sectoral deviations can therefore be interpreted as behavioral or institutional wedges:

$$r_i = r + \varepsilon_i. \quad (145)$$

The mathematical framework does not require any particular psychological mechanism.

24 Psychiatric Analogy and Epistemic Caution

The sign-sector construction should not be interpreted as a model of psychiatric disorders or individual mental states. Mathematical multi-state systems can provide abstract metaphors for state-dependent behavior, but clinical constructs require independent psychological and medical evidence.

A general state-space representation is

$$s_{t+1} = P(s_{t+1} \mid s_t, u_t), \quad (146)$$

which is mathematically distinct from clinical diagnosis.

Accordingly, the present framework uses “sector” and “state” in their mathematical and economic senses.

25 Philosophical Interpretation

The distinction between sector representation and invariant rate raises an epistemological question:

$$\text{Does a representation uniquely determine an underlying object?} \quad (147)$$

Here the answer is no.

Multiple sector representations correspond to the same algebraic r . Hence one must distinguish

$$\text{representation} \neq \text{invariant}. \quad (148)$$

This is closely related to the general philosophical distinction between ontology and representation: a model may admit several descriptions of an object without implying several underlying objects.

26 Religious and Metaphorical Interpretation

The four-sector structure can also be used metaphorically to discuss the difference between multiplicity of representation and unity of an underlying quantity.

One may write schematically

$$\text{many representations} \longrightarrow \text{one invariant}. \quad (149)$$

Such interpretations belong to comparative philosophy and theology rather than to the mathematical proof structure. The equations themselves make no religious claim.

27 Political Science and Institutional Economics

A four-sector economy may contain distinct institutional environments:

$$I_i = \{\text{legal rules, taxes, regulation, liquidity, institutions}\}. \quad (150)$$

Even if the structural benchmark is common,

$$r_i^* = r, \quad (151)$$

actual rates can differ:

$$r_i = r + \varepsilon_i(I_i). \quad (152)$$

This provides a natural institutional interpretation of the deviation term.

28 Geopolitics and International Relations

In an international financial system, sectors could represent jurisdictions, currency blocs, financial centers, or institutional regimes.

Let

$$r_{ij} \quad (153)$$

denote the effective funding or return differential between jurisdictions i and j .
Capital-flow pressure may be modeled as

$$F_{ij} = \lambda_{ij} [r_i - r_j - \kappa_{ij}]_+. \quad (154)$$

The term κ_{ij} may include

$$\begin{array}{ccccccc} \textit{capital} & \textit{currency} & \textit{transaction} & \textit{political} & & \textit{regulatory} & \\ \textit{controls}' & \textit{risk} & \textit{costs} & \textit{risk} & , & \textit{sanctions}, & \textit{barriers} \end{array}. \quad (155)$$

The model therefore provides a compact representation of international financial segmentation.

29 Diplomacy

A diplomatic system can be modeled as a constrained network

$$G = (V, E), \quad (156)$$

where V contains actors and E contains bilateral relationships.

A rate differential may become one component of a broader state vector:

$$\mathbf{d}_{ij} = \begin{pmatrix} \Delta r_{ij} \\ \Delta \pi_{ij} \\ \Delta y_{ij} \\ \Delta q_{ij} \\ \Delta \ell_{ij} \end{pmatrix}. \quad (157)$$

Diplomatic institutions can alter the effective friction

$$\kappa_{ij} \quad (158)$$

through agreements, coordination, guarantees, or restrictions.

30 Warfare Economics

The phrase “sector attack” requires careful interpretation.

In financial terms, an attack can mean

$$\text{capital reallocation} + \text{short selling} + \text{funding withdrawal} + \text{liquidity pressure}. \quad (159)$$

A financial attack is therefore represented by an increase in

$$A_{ij}(t). \quad (160)$$

A literal military conflict is a different phenomenon and cannot be identified with the financial arbitrage process.

A generalized strategic state vector could be written

$$\mathbf{s}_t = \begin{pmatrix} r_t \\ L_t \\ K_t \\ M_t \\ D_t \end{pmatrix}, \quad (161)$$

where the components can represent financial, liquidity, capital, monetary, and defense-related states. Any real-world strategic model would require empirical data and a separate causal framework.

31 Network Representation

The four sectors form a complete graph

$$K_4. \quad (162)$$

There are

$$\binom{4}{2} = 6 \quad (163)$$

undirected pairwise relationships.

The rate-differential matrix is

$$\Delta R = \begin{pmatrix} 0 & r_1 - r_2 & r_1 - r_3 & r_1 - r_4 \\ r_2 - r_1 & 0 & r_2 - r_3 & r_2 - r_4 \\ r_3 - r_1 & r_3 - r_2 & 0 & r_3 - r_4 \\ r_4 - r_1 & r_4 - r_2 & r_4 - r_3 & 0 \end{pmatrix}. \quad (164)$$

It satisfies

$$\Delta R^\top = -\Delta R. \quad (165)$$

The corresponding gross arbitrage-flow matrix is

$$A_{ij} = \lambda_{ij} [\Delta R_{ij} - \kappa_{ij}]_+. \quad (166)$$

32 A Potential-Function Interpretation

Suppose arbitrage capital $K_i(t)$ evolves according to

$$\dot{K}_i(t) = \sum_{j \neq i} [A_{ij}(t) - A_{ji}(t)]. \quad (167)$$

If rates respond to capital flows,

$$\dot{r}_i = -\alpha_i \dot{K}_i + u_i, \quad (168)$$

then capital tends to compress rate differentials under suitable parameter restrictions.

A stylized Lyapunov function is

$$V(t) = \frac{1}{2} \sum_{i < j} [r_i(t) - r_j(t)]^2. \quad (169)$$

An equilibrium condition is

$$V(t) = 0 \quad (170)$$

if and only if

$$r_1 = r_2 = r_3 = r_4. \quad (171)$$

This provides a mathematical route from arbitrage pressure to convergence.

33 Computer Science Representation

The model can be represented as a constraint-satisfaction problem.

Given

$$(a, b, c), \quad (172)$$

the algorithm computes

$$r = \frac{a+b}{2} - \frac{1}{4} \left(c^2 + \frac{(a-b)^2}{c^2} \right). \quad (173)$$

The following pseudocode describes the procedure.

```

INPUT: a, b, c

IF c == 0:
    IF a == b:
        RETURN r = a
    ELSE:
        RETURN infeasible

r = (a + b)/2
  - c^2/4
  - (a - b)^2/(4*c^2)

x = sqrt(a - r)
y = sqrt(b - r)

FOR each sector (s1, s2):
    IF s1*x + s2*y == c:
        mark sector as feasible
    ELSE:
        mark sector as infeasible

RETURN r and feasible sectors

```

The computational complexity of evaluating the four sectors is

$$O(1) \quad (174)$$

per observation.

For a time series of T observations, the complexity is

$$O(T). \quad (175)$$

34 Data Science Pipeline

A practical empirical pipeline is

$$\begin{array}{ccccccc} \text{raw} & \rightarrow & \text{cleaning} & \rightarrow & \text{alignment} & \rightarrow & \text{parameter} \\ \text{data} & & & & & & \text{estimation} \end{array} \rightarrow \begin{array}{c} \text{structural} \\ \text{rate} \end{array} \rightarrow \begin{array}{c} \text{sector} \\ \text{classification} \end{array} \rightarrow \begin{array}{c} \text{arbitrage} \\ \text{diagnostics} \end{array}. \quad (176)$$

Candidate diagnostics include

$$\hat{r}_t = g(\hat{a}_t, \hat{b}_t, \hat{c}_t), \quad (177)$$

$$\hat{\varepsilon}_t = c_t - \sqrt{a_t - \hat{r}_t} - \sqrt{b_t - \hat{r}_t}, \quad (178)$$

$$\widehat{\Delta}r_{ij,t} = \hat{r}_{i,t} - \hat{r}_{j,t}. \quad (179)$$

The residuals can be tested for

$$\text{autocorrelation,} \quad \text{heteroskedasticity,} \quad \text{structural breaks,} \quad \text{non-normality.} \quad (180)$$

35 Machine Learning Extension

A machine-learning model can be used to estimate the sectoral deviations

$$\varepsilon_{i,t} = f_{\theta}(\mathbf{X}_t, i) + u_{i,t}, \quad (181)$$

where $\mathbf{X}_t$ contains market and macroeconomic features.

The structural rate remains

$$r_t = g(a_t, b_t, c_t), \quad (182)$$

while machine learning estimates

$$\hat{r}_{i,t} = \hat{r}_t + \hat{\varepsilon}_{i,t}. \quad (183)$$

This separation is desirable because it prevents the machine-learning model from unnecessarily re-learning an analytically known structural relation.

36 Reinforcement Learning Interpretation

A sector can be modeled as an agent choosing capital allocation $u_{i,t}$.

The state is

$$\mathbf{s}_t = (r_{1,t}, \dots, r_{4,t}, K_{1,t}, \dots, K_{4,t}, \mathbf{X}_t). \quad (184)$$

The agent maximizes

$$J_i = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t U_i(\mathbf{s}_t, u_{i,t}) \right]. \quad (185)$$

The resulting policy is

$$u_{i,t} = \pi_i(\mathbf{s}_t). \quad (186)$$

A multi-agent system can therefore learn endogenous capital-flow dynamics around the radical-balance benchmark.

37 Causal Inference

A central empirical question is whether a sectoral rate differential causes capital movement.

Let

$$Y_{ij,t} \tag{187}$$

represent a capital-flow outcome and

$$D_{ij,t} = [r_{i,t} - r_{j,t} - \kappa_{ij,t}]_+. \tag{188}$$

A basic causal model is

$$Y_{ij,t} = \alpha + \tau D_{ij,t} + \beta^\top \mathbf{X}_{ij,t} + u_{ij,t}. \tag{189}$$

Identification requires assumptions concerning confounding, simultaneity, measurement, and selection. The radical-balance equation alone does not identify τ .

38 Structural Breaks

Because the sector classification depends on

$$c^2 \quad \text{and} \quad |a - b|, \tag{190}$$

sector transitions occur at

$$c^2 = |a - b|. \tag{191}$$

This is therefore a natural structural-break boundary.

A regime indicator can be defined as

$$S_t = \begin{cases} ++, & c_t^2 > |a_t - b_t|, \\ +-, & a_t - b_t > c_t^2, \\ -+, & b_t - a_t > c_t^2. \end{cases} \tag{192}$$

This creates a discrete regime process embedded in a continuous rate process.

39 A Geometric Interpretation

The original radical equation can be interpreted geometrically.

Let

$$x = \sqrt{a - r}, \quad y = \sqrt{b - r}. \tag{193}$$

Then the ++ sector is

$$x + y = c, \tag{194}$$

which is a line in the first quadrant.

The three additional sectors correspond to

$$x - y = c, \tag{195}$$

$$-x + y = c, \tag{196}$$

$$-x - y = c. \tag{197}$$

These are the four sign-related line sectors.

The transformation from x, y to r is

$$r = a - x^2 = b - y^2. \tag{198}$$

Thus the system combines linear sector geometry in radical coordinates with nonlinear projection onto the interest-rate coordinate.

$$\sqrt{a(t) - r(t)} + \sqrt{b(t) - r(t)} = c(t) \quad \sqrt{a(t) - r(t)} - \sqrt{b(t) - r(t)} = c(t)$$

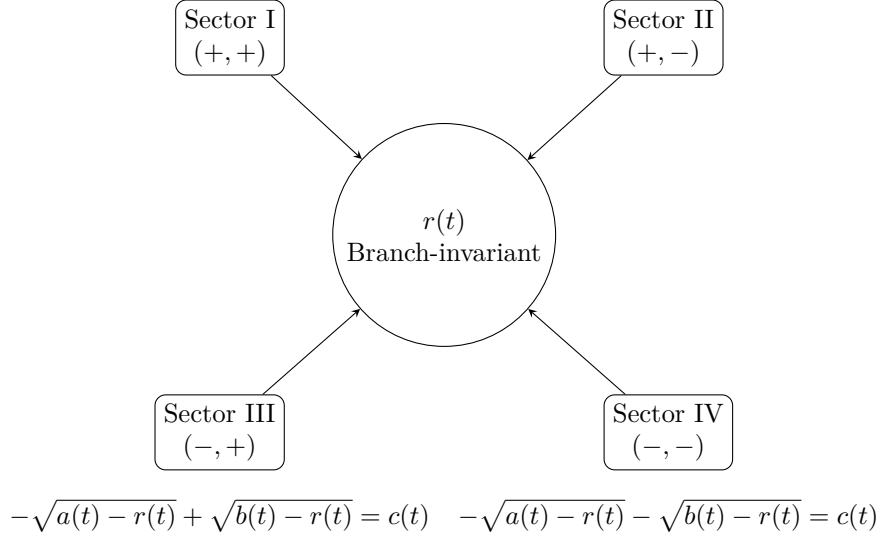


Figure 1: The four-sector radical-balance system. Each sector is generated by a distinct sign configuration of the two radical terms, while the interest rate $r(t)$ remains branch-invariant.

40 Physical and Financial Stability

The structural rate is stable with respect to small perturbations in a, b, c when the gradient

$$\nabla g(a, b, c) \tag{199}$$

is bounded.

Near $c = 0$, however,

$$\frac{\partial r}{\partial c} = -\frac{c}{2} + \frac{(a - b)^2}{2c^3} \tag{200}$$

can become very large.

This corresponds mathematically to a region of high sensitivity.

In financial applications, such sensitivity can be interpreted as a warning that estimation error in c may be strongly amplified.

41 Risk Interpretation

Define the structural rate error

$$e_t = r_t^m - r_t^{\text{RB}}, \tag{201}$$

where r_t^m is an observed market rate and

$$r_t^{\text{RB}} = \frac{a_t + b_t}{2} - \frac{c_t^2}{4} - \frac{(a_t - b_t)^2}{4c_t^2}. \tag{202}$$

Then

$$e_t \tag{203}$$

can be studied as a pricing, liquidity, credit, or model-risk residual.

A volatility model could take the form

$$e_t = \sigma_t z_t, \quad (204)$$

with

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (205)$$

This connects the structural rate model to conditional-volatility econometrics.

42 Systemic Risk Extension

Let

$$\mathbf{A}_t = [A_{ij,t}] \quad (206)$$

be the arbitrage-flow matrix.

A network systemic-risk indicator may be defined as

$$\mathcal{R}_t = \rho(\mathbf{A}_t), \quad (207)$$

where

$$\rho(\mathbf{A}_t) \quad (208)$$

denotes the spectral radius.

This quantity can measure the intensity of network feedback in a stylized model, although its empirical interpretation requires calibration.

43 Four-Sector Stability Theorem

Consider a linearized sectoral adjustment model

$$\dot{\mathbf{r}} = -\Gamma L \mathbf{r} + \mathbf{u}, \quad (209)$$

where L is the graph Laplacian of the four-sector network and Γ is a positive diagonal matrix.

The Laplacian satisfies

$$L \mathbf{1} = 0. \quad (210)$$

Hence the common-rate vector

$$\mathbf{r} = r \mathbf{1} \quad (211)$$

is an equilibrium direction.

If the graph is connected and adjustment coefficients are positive, all non-common components are damped under the homogeneous dynamics.

This formalizes the intuitive statement that arbitrage pressure tends to eliminate pure rate differentials in a frictionless connected network.

44 A Unified Four-Sector Architecture

The complete model can therefore be represented as

$$\begin{array}{c}
 a(t), b(t), c(t) \\
 \downarrow \\
 \text{Radical-Balance Equation} \\
 \downarrow \\
 r(t) \\
 \downarrow \\
 \text{Sectoral Perturbations} \\
 \downarrow \\
 r_i(t) = r(t) + \varepsilon_i(t) \\
 \downarrow \\
 \text{Rate Differentials} \\
 \downarrow \\
 \text{Arbitrage Flows} \\
 \downarrow \\
 \text{Capital Reallocation} \\
 \downarrow \\
 \text{Dynamic Rate Adjustment.}
 \end{array} \tag{212}$$

This decomposition separates structural determination from endogenous market dynamics.

45 Interdisciplinary Synthesis

The framework admits several levels of interpretation.

Table 2: Interdisciplinary interpretation of the four-sector system

Field	Mathematical object	Possible interpretation
Economics	$r(t)$	Structural interest-rate benchmark
Finance	$r_i - r_j$	Arbitrage opportunity differential
Econometrics	$\hat{r}_t$	Structural estimator
Statistics	∇g	Sensitivity and uncertainty propagation
Physics	Sign symmetry	Discrete symmetry analogy
Chemistry	Sector states	Multiple representations of an observable
Biology	State transitions	Regime-switching analogy
Psychology	ε_i	Heterogeneous decision wedges
Philosophy	Invariant r	Representation versus object
Religion	Sector unity	Metaphorical multiplicity/unity analogy
Political science	I_i	Institutional sector differences
Geopolitics	κ_{ij}	Cross-border financial frictions
Diplomacy	Network G	Bilateral institutional relations
Welfare economics	W	Distributional consequences
Warfare economics	A_{ij}	Financial pressure metaphor
Computer science	Algorithm	Constraint-solving representation
Data science	Pipeline	Empirical implementation
AI/ML	f_θ	Learned sectoral deviations
Network science	L	Systemic adjustment structure

These interpretations should not be conflated. The theorem-level claims are mathematical. The economic claims require economic assumptions. The interdisciplinary extensions are modeling analogies or research hypotheses unless empirically validated.

46 Empirical Research Program

A complete empirical investigation could proceed through the following stages.

1. Construct time series for a_t , b_t , and c_t .

2. Compute the structural rate using

$$\hat{r}_t = \frac{\hat{a}_t + \hat{b}_t}{2} - \frac{\hat{c}_t^2}{4} - \frac{(\hat{a}_t - \hat{b}_t)^2}{4\hat{c}_t^2}. \quad (213)$$

3. Classify the feasible sector using

$$c_t^2 \quad \text{versus} \quad |a_t - b_t|. \quad (214)$$

4. Compare the structural rate with observed market rates.

5. Estimate sectoral deviations

$$\varepsilon_{i,t} = r_{i,t}^m - \hat{r}_t. \quad (215)$$

6. Construct arbitrage-pressure variables

$$A_{ij,t} = \lambda_{ij} [r_{i,t}^m - r_{j,t}^m - \kappa_{ij,t}]_+. \quad (216)$$

7. Test whether rate differentials predict subsequent capital flows.

8. Estimate dynamic feedback between capital flows and rate differentials.

9. Test for structural breaks at

$$c_t^2 = |a_t - b_t|. \quad (217)$$

10. Conduct robustness analysis across frequencies, samples, jurisdictions, and alternative definitions of a , b , and c .

47 Research Hypotheses

The mathematical framework generates several empirically testable hypotheses.

47.1 Structural hypothesis

$$H_1 : \quad r_t^m \approx g(a_t, b_t, c_t). \quad (218)$$

47.2 Arbitrage hypothesis

$$H_2 : \quad r_{i,t} - r_{j,t} > \kappa_{ij,t} \implies \mathbb{E}_t[\Delta K_{ij,t+1}] > 0. \quad (219)$$

47.3 Convergence hypothesis

$$H_3 : \quad \Delta r_{ij,t} > 0 \implies \mathbb{E}_t[\Delta r_{ij,t+1}] < \Delta r_{ij,t}. \quad (220)$$

47.4 Spectral hypothesis

The nonlinear structure predicts frequency mixing, so

$$H_4 : \quad \text{power at generated harmonics is related to nonlinear variation in } c_t. \quad (221)$$

These are empirical hypotheses rather than mathematical theorems.

48 Limitations

Several limitations must be emphasized.

First, the four-sector radical equations themselves do not generate four different interest rates. They generate one algebraically invariant rate.

Second, arbitrage cannot be inferred solely from the existence of different sign sectors. It requires economically distinct sectoral rates or wedges.

Third, the term “attack” is metaphorical in the financial formulation. Actual military conflict is governed by different state variables and causal mechanisms.

Fourth, the structural estimator can be unstable when c_t approaches zero.

Fifth, the mapping from a , b , and c to economic observables is not uniquely determined by the mathematics.

Sixth, interdisciplinary analogies do not constitute evidence for claims in psychology, psychiatry, biology, chemistry, political science, or international relations.

49 Conclusion

The four-sector radical-balance system begins with

$$s_1 \sqrt{a(t) - r(t)} + s_2 \sqrt{b(t) - r(t)} = c(t), \quad s_1, s_2 \in \{-1, +1\}. \quad (222)$$

Its central mathematical result is

$$r(t) = \frac{a(t) + b(t)}{2} - \frac{1}{4} \left[c(t)^2 + \frac{[a(t) - b(t)]^2}{c(t)^2} \right]. \quad (223)$$

The four sign sectors therefore constitute different radical-coordinate representations of a common algebraic interest-rate object.

The principal-square-root constraints determine which sector is feasible. For $c(t) > 0$, the sector boundary is

$$c(t)^2 = |a(t) - b(t)|. \quad (224)$$

The same condition identifies the extremal rate

$$r_{\max}(t) = \min\{a(t), b(t)\}. \quad (225)$$

In the exact model,

$$r_{++} = r_{+-} = r_{-+} = r_{--}, \quad (226)$$

so the system is arbitrage neutral.

The economically richer model introduces sectoral deviations,

$$r_i(t) = r(t) + \varepsilon_i(t), \quad (227)$$

which produce rate differentials

$$\Delta r_{ij}(t) = \varepsilon_i(t) - \varepsilon_j(t). \quad (228)$$

These differentials can generate arbitrage pressure,

$$A_{ij}(t) = \lambda_{ij} [\Delta r_{ij}(t) - \kappa_{ij}(t)]_+. \quad (229)$$

Thus the complete conceptual architecture is

$$\boxed{\begin{array}{ccccccc} \text{radical} & \rightarrow & \text{branch-invariant} & \rightarrow & \text{sectoral} & \rightarrow & \text{capital} & \rightarrow & \text{dynamic} \\ \text{balance} & & \text{benchmark} & & \text{deviations} & \rightarrow & \text{arbitrage} & \rightarrow & \text{convergence} \\ & & & & & & \text{flows} & & \end{array}} \quad (230)$$

The principal contribution of the framework is therefore not merely the closed-form solution for $r(t)$. It is the separation of an invariant structural benchmark from the sector-specific deviations that can generate financial disequilibrium, arbitrage, and network dynamics.

A Complete Sector Table

Table 3: Four radical sectors

Sector	s_1	s_2	Radical equation	Algebraic r
++	+1	+1	$x + y = c$	$g(a, b, c)$
+-	+1	-1	$x - y = c$	$g(a, b, c)$
-+	-1	+1	$-x + y = c$	$g(a, b, c)$
--	-1	-1	$-x - y = c$	$g(a, b, c)$

where

$$g(a, b, c) = \frac{a+b}{2} - \frac{c^2}{4} - \frac{(a-b)^2}{4c^2}. \quad (231)$$

B Degenerate Case $c = 0$

The closed-form expression contains c^{-2} , so $c = 0$ must be treated separately.

The original equation becomes

$$s_1\sqrt{a-r} + s_2\sqrt{b-r} = 0. \quad (232)$$

For the ++ sector,

$$\sqrt{a-r} + \sqrt{b-r} = 0 \quad (233)$$

requires

$$a - r = 0, \quad b - r = 0, \quad (234)$$

and hence

$$\boxed{a = b = r.} \quad (235)$$

Other sectors can admit additional boundary configurations depending on the relationship between a and b . Thus the $c = 0$ case should be handled directly from the original equations rather than by taking a formal limit of the closed-form expression.

C Useful Differential Identities

For

$$r(a, b, c) = \frac{a+b}{2} - \frac{c^2}{4} - \frac{(a-b)^2}{4c^2}, \quad (236)$$

the partial derivatives are

$$r_a = \frac{1}{2} - \frac{a-b}{2c^2}, \quad (237)$$

$$r_b = \frac{1}{2} + \frac{a-b}{2c^2}, \quad (238)$$

$$r_c = -\frac{c}{2} + \frac{(a-b)^2}{2c^3}. \quad (239)$$

The identity

$$r_a + r_b = 1 \quad (240)$$

is particularly useful.

It shows that a common infinitesimal increase in both anchors produces, holding c fixed,

$$dr = da = db \quad (241)$$

when

$$da = db. \quad (242)$$

D Algorithm for Empirical Implementation

Algorithm RadicalBalanceFourSector

Input:

time series $a[t]$, $b[t]$, $c[t]$

For each t :

If $c[t] == 0$:

 solve original four equations directly
 continue

$r[t] =$

$(a[t] + b[t])/2$
 $- c[t]^2/4$
 $- (a[t] - b[t])^2/(4*c[t]^2)$

$x[t] = \sqrt{a[t] - r[t]}$

$y[t] = \sqrt{b[t] - r[t]}$

sector[1] = (+1, +1)

sector[2] = (+1, -1)

sector[3] = (-1, +1)

sector[4] = (-1, -1)

For each sector ($s1$, $s2$):

 residual =
 $s1*x[t] + s2*y[t] - c[t]$

 If residual is numerically zero:

 sector is feasible

 Else:

 sector is infeasible

Store:

$r[t]$

 feasible sector

$x[t]$

$y[t]$

Return complete time series

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Glossary

Arbitrage A strategy exploiting economically equivalent or sufficiently similar positions whose net return differential exceeds the relevant costs and risks.

Branch A mathematically distinct solution representation generated by a choice of sign, root, or other multivalued operation.

Branch invariance The property that an observable quantity remains unchanged under a family of alternative mathematical representations.

Four-sector radical-balance system The four equations obtained by independently assigning positive or negative coefficients to the two radicals.

Frictions Transaction costs, taxes, liquidity costs, regulatory barriers, funding constraints, currency risk, and other impediments to arbitrage.

Interest-rate benchmark The structurally determined rate $r(t)$ generated by the radical-balance equation.

Principal square root The nonnegative square root of a nonnegative real number.

Radical coordinate A variable such as

$$x(t) = \sqrt{a(t) - r(t)} \tag{243}$$

or

$$y(t) = \sqrt{b(t) - r(t)}. \tag{244}$$

Radical-balance interest rate The rate determined by the relation

$$\sqrt{a-r} + \sqrt{b-r} = c \quad (245)$$

and its sign-sector extensions.

Sector One of the four sign configurations

$$(+, +), \quad (+, -), \quad (-, +), \quad (-, -). \quad (246)$$

Sectoral deviation The difference between an observed or modeled sectoral rate and the common structural benchmark:

$$\varepsilon_i = r_i - r. \quad (247)$$

Structural identification Identification obtained from the mathematical structure of a model rather than from a statistical estimation procedure.

Frequency mixing Generation of new spectral components by a nonlinear transformation of signals.

Systemic risk The possibility that disturbances propagate through interconnected agents, markets, or institutions.

No-arbitrage equilibrium A state in which relevant risk-adjusted return differentials do not exceed the costs and constraints required to exploit them.

The End